EVA Engineering Spec — v0.3 Research Node

Stage 1 output — Architect Agent
Status: DRAFT — pending `core.py` confirmation (see Open Questions)

Legend

- ✅ **CONFIRMED** — verified from actual running code (`api.py`)
- 📋 **PLANNED** — from the v0.3 Engineering Handoff Package, not yet verified in code
- ⚠️ **CONFLICT** — handoff doc says one thing, real code does another (real code wins per the One Rule)
- ❓ **OPEN** — unknown until `core.py` (and `tools/gateway.py`, `models.py`) are provided

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1. Product Target

✅ EVA Research Node: a private research assistant with controlled, gated agents — not full autonomy.

Confirmed end-to-end flow (matches both `api.py` routes and the handoff doc):

1. User submits a mission (objective, optional source text or sources)
2. Pixie converts the request into a structured goal *(❓ not yet seen in code — only referenced in handoff doc)*
3. EVA Core creates task(s)
4. Sentinel authorizes tool/task usage (`core.sentinel.review_task`)
5. Nova researches using Tool Gateway (`web_search`, confirmed via `/tool/execute`, `/tool/list`)
6. Ghost stores validated knowledge (`core.ghost.ingest`, `core.ghost.search`)

7. EVA produces a final report (`mission.report`)

2. Runtime Stack

Layer	Handoff doc says (📋)	Actual code (✅)
Backend framework	FastAPI	⚠️ stdlib `http.server` (`ThreadingHTTPServer`) — deliberate, per `api.py` docstring: sandbox has no network access to install packages. Handler bodies are written to be portable to FastAPI later.
Database	PostgreSQL	❓ Unknown — `main()` takes a `db_path` argument (default `eva.db`), which is suggestive of SQLite but not confirmed without `core.py` / `memory/database.py`.
Memory/vector search	pgvector	❓ Unconfirmed. `ghost.search(query, mission_id)` exists but its retrieval method (keyword vs. vector) is unknown.
Agent framework	LangGraph / OpenAI Agents SDK	❓ No evidence yet of either in confirmed code. Agents (`Pixie`, `Nova`, `Sentinel`, `Ghost`) appear to be referenced as attributes/modules on `EVACore`, not via a third-party orchestration framework.
Frontend	React	⚠️ Actual deployed frontend is a single-file vanilla HTML/JS dashboard (no build step, no React) — confirmed from uploaded `index.html`.
Deployment	Docker Compose	❓ Unconfirmed — no Dockerfile or compose file has been provided yet.

Layer	Handoff doc says (📋)	Actual code (✅)
Entrypoint	—	✅ `python -m eva.api [port] [db_path]` — default port `8000`, default db `eva.db`

Guidance for Stage 2+: treat the "Handoff doc says" column as background intent only. Do not build against it where the "Actual code" column has a ✅ or ⚠️ entry — that reflects what's already running.

3. Repository Layout

📋 Proposed in handoff doc (not yet verified against a real file tree):

```
EVA/
  app/
    main.py
    core/ (eva.py, mission.py, planner.py)
    agents/ (pixie.py, nova.py, sentinel.py, ghost.py)
    memory/ (database.py, retrieval.py)
    security/ (permissions.py)
    tools/ (search.py, documents.py)
  tests/
  docs/
  docker-compose.yml
  README.md
```

✅ Confirmed to actually exist (from file status table + api.py imports):

```
eva/
  api.py                (confirmed, full content on file)
  core.py                <- imported as `from eva.core import
EVACore` — NOT YET SEEN
  models.py              <- imported as `from eva.models
import ...` — NOT YET SEEN
  tools/
    __init__.py          (exists per status table)
    base.py               (exists per status table)
    web_search.py         (exists — DuckDuckGo, no API key,
per status table)
```

```
    gateway.py              (exists — "Sentinel-gated tool
execution", per status table)
```

❓ Open: whether `eva/agents/`, `eva/memory/`, `eva/security/` exist as separate modules, or whether Pixie/Nova/Sentinel/Ghost logic lives inside `core.py` and `tools/gateway.py`. `api.py` only ever calls `core.sentinel` and `core.ghost` — it never imports Pixie or Nova directly, which suggests those may be internal to `EVACore`, not top-level agent files. **This needs verification before Stage 2 scaffolds a directory tree.**

4. Agent Rules

📋 From handoff doc, not yet cross-checked against `core.py`:

Agent	Input	Output	Constraints
Pixie	Human request	Mission object	No system access
Nova	Approved research task	`{Finding, Evidence, Confidence, Unknowns}`	Must go through Tool Gateway (✅ confirmed pattern: `/tool/execute` routes through Sentinel-gated gateway)
Sentinel	Task	Authorization decision	Must approve tool use, external access, sensitive ops. Cannot modify EVA rules. ✅ Confirmed live: `core.sentinel.review_task(task, user_confirmed)` → `{decision, auth_level, reason}`
Ghost	Validated statement	Memory record	Stores `{content, source, confidence, timestamp, classification}`. ✅ Confirmed live: `core.ghost.ingest(mission_id, statement, source_id)` → record + conflict list; `core.ghost.search(query, mission_id)`

5. Explicit Non-Goals (carried forward from handoff doc — still binding)

No sentience claims, no unrestricted autonomy, no self-modification, no consciousness layers, no large multi-agent swarms. These remain out of scope for the Research Node target.

6. Open Questions Blocking Full Confidence

1. What does `core.py` actually contain — does `EVACore.__init__` take `db_path`, and how are Pixie/Nova wired in?
2. What database engine backs `eva.db` — SQLite (suggested by filename) or something else?
3. What's inside `models.py` — full enum values for `AgentName`, `AuthLevel`, `TaskStatus`, and the `Mission` / `User` / `MemoryRecord` field definitions?
4. What does `tools/gateway.py` actually enforce — is Sentinel review mandatory for every tool call, or optional per `auth_level`?

Recommendation: Stage 2 (Database/Backend Agent) should not begin schema migration work until Q1–Q3 above are resolved, to avoid re-deriving a schema that conflicts with what `core.py` already implements.